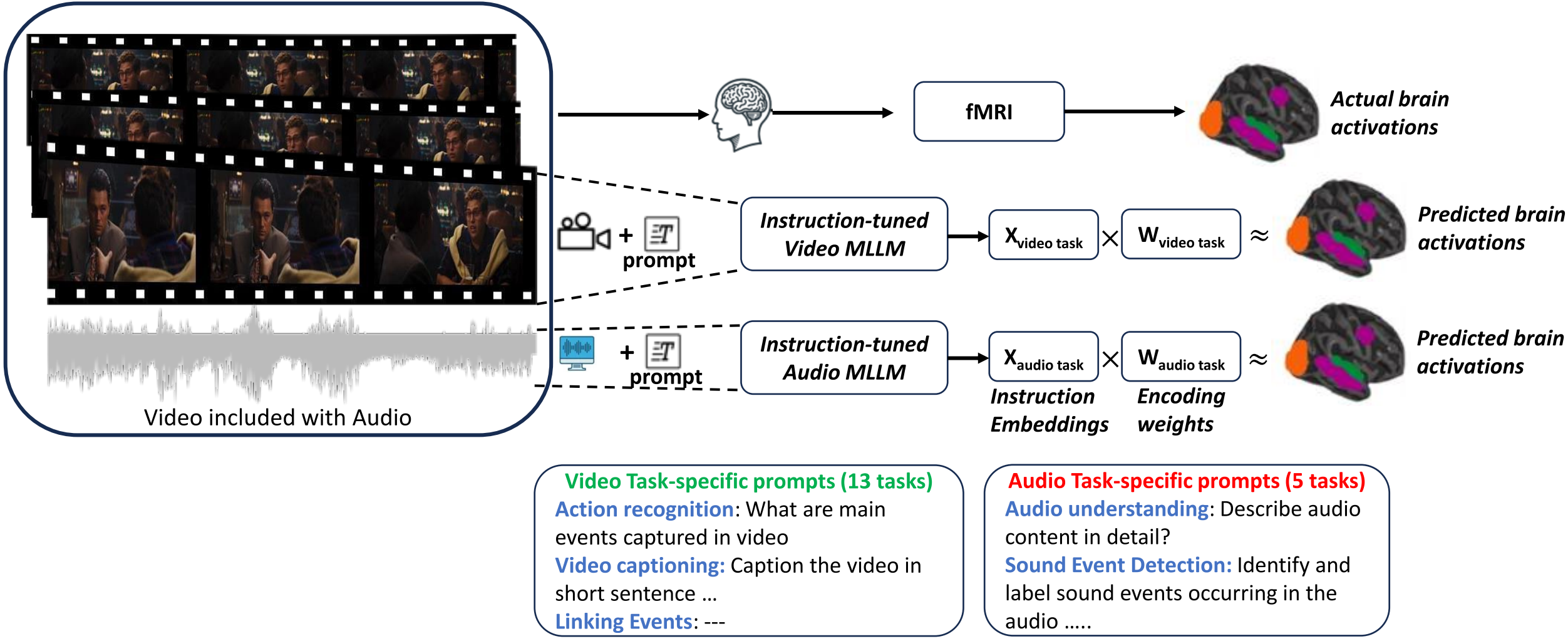
Video included with Audio
+ prompt
+ prompt
fMRI
Actual brain activations
Instruction-tuned Video MLLM
Instruction-tuned Audio MLLM
X video task
W video task
X audio task
W audio task
Instruction Embeddings
Encoding weights
Predicted brain activations
Predicted brain activations
Video Task-specific prompts (13 tasks)
Action recognition: What are main events captured in video
Video captioning: Caption the video in short sentence …
Linking Events: ---
Audio Task-specific prompts (5 tasks)
Audio understanding: Describe audio content in detail?
Sound Event Detection: Identify and label sound events occurring in the audio …..